jambronotes.sty

Feature Demo

v1.1-dev — *June 29, 2026*

Contents

List of Boxes

I Typography and Fonts

I.A Body Text

Body text is set in **Source Sans Pro** (loaded via `[default]{sourcesanspro}` as the document default), while section headings use **Fira Sans** (the `\firatitle` switch) for contrast. Captions follow the body font. Math is rendered in the standard Computer Modern math fonts.

A short paragraph: The quick brown fox jumps over the lazy dog. Numbers: 0123456789. Greek: $\alpha, \beta, \gamma, \delta, \epsilon$. Operators: $\sum_{i=1}^{n} x_i$, $\int_0^\infty e^{-x}\,dx$.

I.B Handwriting Font

The Augie font is available for hand-drawn effects. `\hand` is a font switch (use inside a group `{...}`); `\handbold` takes its text as an argument:

```
{\hand This is the Augie handwriting font.}
\handbold{This is handbold (default stroke 0.6).}
\handbold[0.3]{Lighter stroke.}   % optional width argument
\handbold[1.0]{Heavier stroke.}
```

renders as:

- This sentence is set in the Augie handwriting font.
- **This sentence uses handbold — bold Augie via PDF stroke literal.**
- Stroke width is an optional argument: light (0.2), **default (0.6)**, **heavy (1.1)**.
- Mixed: regular text alongside handwritten words in the same line.

I.C Pencil Underlines

Default underline uses the `carandache` color (goldenrod). The color is passed as an optional argument:

```
\lapis{default underline}
\lapis[tomato]{custom color}
```

The optional argument accepts any defined color (code → result):

- `\lapis{carandache gold}` → carandache gold (default)
- `\lapis[tomato]{tomato red}` → tomato red
- `\lapis[note]{note blue}` → note blue
- `\lapis[cobalt]{cobalt teal}` → cobalt teal
- `\lapis[BoEpurple]{BoE purple}` → BoE purple

Underlines work inside longer sentences too: the model predicts that output falls whenever the borrowing constraint binds.

II Pencil Boxes

II.A Inline Boxes

Default gold box uses no options. The color is set with the `color=` key:

```
\lapisbox{default gold box}
\lapisbox[color=tomato]{custom color}
```

Color variants (code → result):

- `\lapisbox{gold}` → $\boxed{\text{gold}}$ (default)
- `\lapisbox[color=tomato]{tomato}` → $\boxed{\text{tomato}}$
- `\lapisbox[color=note]{note blue}` → $\boxed{\text{note blue}}$
- `\lapisbox[color=cobalt]{cobalt}` → $\boxed{\text{cobalt}}$
- `\lapisbox[color=BoEpurple]{BoE purple}` → $\boxed{\text{BoE purple}}$

Boxes work mid-sentence: the key result is that $\boxed{\beta > 0}$ whenever agents are impatient.

II.B Equation Boxes

`\lapisboxeq` wraps display math in a pencil box without requiring the caller to enter math mode. Put the math content directly in the argument (no $ or `\[`):

```
\[
    \lapisboxeq{Y_t = C_t + I_t + G_t + NX_t}
    \qquad
    \lapisboxeq[color=tomato]{r_t = \rho + \phi_\pi \pi_t}
\]
```

renders as:

$$\boxed{Y_t = C_t + I_t + G_t + NX_t} \qquad \boxed{r_t = \rho + \phi_\pi(\pi_t - \bar{\pi}) + \phi_y \tilde{y}_t}$$

It also works inline: `\lapisboxeq{r = \rho}` → the steady-state condition is $\boxed{r = \rho}$.

II.C Labeled boxes and pencil arrows

Boxes with `label=` create named TikZ nodes. A later `tikzpicture` (with the `tpstyle` option) targets them by name; compile **twice** for cross-picture references to resolve:

```
The \lapisbox[color=tomato,label=firesale]{firesale} externality
\begin{tikzpicture}[tpstyle]
    \draw[pencil,->,tomato!80] ([yshift=-2pt]firesale.south east)
        to[bend right] +(+0.7,-0.5)
        node[anchor=west,opacity=1,tomato] {\hand A comment};
\end{tikzpicture}
```

renders as:

- The $\boxed{\text{aggregate demand}}$ externality
- The $\boxed{\text{firesale}}$ externality

A comment

III Annotations and Notes

III.A Inline Todo Notes

```
\NB{remember to add the proof here}
```

renders as:

This is an inline note produced by the `NB` command. It uses the `note` blue background from `todonotes`.

Todo notes are useful for flagging gaps during drafting. They sit in the text flow (not the margin) and are easy to search.

III.B Draft Annotations

The default user and color are set once in the preamble; individual calls override them with `user=` and `color=` keys:

```
% in the preamble:
\renewcommand{\memouser}{ACB}
\renewcommand{\memocolor}{BoEred}
% in the body:
\memo{uses the preamble defaults}
\memo[user=JB,color=note]{override both}
\memo[color=cobalt]{override color only}
```

renders as:

ACB: Default user (ACB, set once in the preamble) and default color (BoEred).

JB: Custom user and color via optional key=value argument.

ACB: Color-only override — username still comes from `\memouser`.

Regular text continues after the annotation without extra spacing.

III.C Note Boxes

A `notebox` is a breakable shaded callout. It is an environment, so wrap content between `\begin{notebox}` and `\end{notebox}`:

```
\begin{notebox}
    \textbf{Key result.} Some highlighted text.
\end{notebox}
```

renders as:

Key result. In the Korinek–Simsek (2016) model, competitive borrowers do not internalise that their deleveraging reduces aggregate income. The social planner internalises this and chooses a lower debt level in period 0.

Note boxes accept arbitrary content including math and itemize:

Two externalities.
- **Aggregate demand**: deleveraging $\Rightarrow$ lower income $\Rightarrow$ tighter constraint.
- **Firesale**: forced selling $\Rightarrow$ lower asset prices $\Rightarrow$ lower collateral.

Both generate over-borrowing in equilibrium.

Adding `\boxentry{title}` as the first line turns a plain `notebox` into a *numbered* box: it gains a dotted "Box X.Y" header, an entry in the List of Boxes (shown after the table of contents), and a label you can `\ref`. A bare `notebox` (above) stays unnumbered and header-less.

```
\begin{notebox}
    \boxentry{The running example}\label{box:demo}
    Body text of the numbered box ...
\end{notebox}
```

```
As shown in Box~\ref{box:demo} ...
```

renders as:

Box 3.A

The running example. A bivariate VAR(1) keeps every object small enough to write out in full. Box numbers run " arabic-section.alpha " (here 3.A) and reset each section.

Box 3.B

This box increments the letter within the section, so it is Box 3.B. Cross-references resolve because `\boxentry` calls `\refstepcounter`.

Reference them anywhere: the example is in Box 3.A and the follow-up in Box 3.B.

Although a `notebox` and an `\NB` note share the same `note` blue, they serve opposite purposes. A `notebox` is an environment for *permanent* content you want to keep in the finished document — a highlighted result, definition, or remark — and it lightly shades (`note!10`), breaks across pages, and holds a body of any length. `\NB{...}` is a single-argument command for *transient* draft markers ("fix this later"); it renders as a more saturated inline banner (`note!40`) via `todonotes`. Unlike `\memo`, `\NB` has no suppression toggle, so remove such markers before finalising.

IV Code Listings

IV.A The `code` Environment

The `code` environment typesets a verbatim block on a light-gray background. It is deliberately monochrome — all text is black, with no syntax highlighting — and breaks across pages. Content is taken verbatim, so no escaping is needed:

```
\begin{code}
% MATLAB: estimate a reduced-form VAR(1)
[VAR, VARopt] = VARmodel(data, 1);
disp(VAR.Fcomp);    % companion matrix
\end{code}
```

renders as:

```
% MATLAB: estimate a reduced-form VAR(1)
root = fileparts(fileparts(mfilename('fullpath')));
addpath(fullfile(root, 'VAR'));
[VAR, VARopt] = VARmodel(data, 1);    % equation-by-equation OLS
disp(VAR.Fcomp);                      % companion matrix
rmpath(genpath(root));
```

IV.B Inline Code Chips

`\mc` and `\mci` set inline code on the same light-gray background. `\mc` uses a fixed code size; `\mci` follows the surrounding font size:

```
Call \mc{addpath(genpath(root))} to add the toolbox; the
\mci{root} variable holds the install path.
```

renders as: call `addpath(genpath(root))` to add the toolbox; the `root` variable holds the install path.

V Mathematics

V.A Standard Environments

The `\E` shorthand gives the expectations operator (use `\E_t` for a time subscript):

```
\[ \E_t\left[ u'(C_{t+1}) \right] = 1 \]
```

renders as:

$$\mathbb{E}_t\left[\frac{u'(C_{t+1})}{u'(C_t)}\frac{1+r_{t+1}}{1+\pi_{t+1}}\right] = 1$$

A multi-line derivation using `align`:

$$\begin{aligned} C_t^b + B_{t+1} &= y_t + (1+r_t)B_t && (1) \\ B_{t+1} &\leq \phi && (2) \\ \lambda_t &= u'(C_t^b) - \beta^b(1+r_t)\mathbb{E}_t[u'(C_{t+1}^b)] \geq 0 && (3) \end{aligned}$$

VI Tables

VI.A Standard Tabularx

The `Y` column type is a centered auto-width `X` column. The style deliberately redefines `\notesize` to produce compact 8.5pt annotation text below the table:

A publication-style pattern wraps the table in a fixed-width `minipage` (so the caption and note share the table's measure), uses `\shortstack` for two-line headers, and sets a long `\notesize` note that wraps across several lines:

```
\begin{table}[ht!]
    \centering
    \begin{minipage}[b]{.7\textwidth}
        \begin{center}\small
        \begin{tabularx}{\textwidth}{lYY}
        \toprule
            & \shortstack{Demand\\ Shock} & \shortstack{Monetary\\ Policy Shock} \\
            & ($\varepsilon_{t}^{Demand}$) & ($\varepsilon_{t}^{MonPol}$) \\
    \midrule\addlinespace
            Output growth ($y_{t}$)            & + & - \\
            Short-term Interest Rate ($r_{t}$)  & + & + \\ \bottomrule
        \end{tabularx}%
        \end{center}
        \vspace{0.4cm}
        \caption{\scshape{Sign Restrictions for Demand Shock \\ and Monetary Policy Shock}}
        \vspace{0.2cm}
        {\notesize {\scshape Note.} The signs represent restrictions on the
        elements of the structural impact matrix $B$ ...}
        \label{tab:SR}
    \end{minipage}
\end{table}
```

renders as:

Table 1: SIGN RESTRICTIONS FOR DEMAND SHOCK AND MONETARY POLICY SHOCK

	Demand Shock (ε_t^{Demand})	Monetary Policy Shock (ε_t^{MonPol})
Output growth (y_t)	+	-
Short-term Interest Rate (r_t)	+	+

NOTE. The signs represent restrictions on the elements of the structural impact matrix B. '$+$' and '$-$' indicate a positive and negative contemporaneous response, respectively. A demand shock raises both output growth and the short-term interest rate, as monetary policy tightens to contain the expansion. A contractionary monetary policy shock lowers output growth and raises the interest rate, consistent with an unexpected tightening of monetary policy.

VI.B Table with Row Colors

The [table] option on xcolor (loaded by the theme) enables \rowcolor, placed at the start of a row:

```
\rowcolor{note!8} Discount factor & $\beta^b$ & 0.90 \\
```

renders as:

Table 2: MODEL PARAMETERS

Parameter	**Symbol**	**Value**
Discount factor (borrower)	β^b	0.90
Discount factor (lender)	β^l	0.95
Borrowing limit	ϕ	0.75
Frisch elasticity	φ	1.00
Price rigidity	κ	0.50

NOTE. Illustrative calibration of the borrower–lender model used throughout this demo; the values are not estimated and serve only to exercise the \rowcolor shading and the Y column type within the same fixed-width minipage layout as Table 1.

VII Paragraph Numbering

Numbered paragraphs use \para. Place it at the start of the paragraph; the counter increments automatically and typesets a number followed by a quad of space:

```
\para This paragraph is automatically numbered.
\para So is this one, with the next number.
```

renders as:

1. This is the first numbered paragraph. The number is produced by \para and counts independently of section numbers. It is useful for notes that cross-reference specific arguments.

2. This is the second numbered paragraph. The borrowing constraint $B_{t+1} \leq \phi$ is assumed to bind in period 1 but not in period 0.

3. This is the third numbered paragraph. The social planner internalises the aggregate demand externality and imposes a tax τ^* on period-0 borrowing.

Because \para calls \refstepcounter, a numbered paragraph can carry a \label and be cited later with \ref:

```
\para\label{para:key} The constraint binds in period 1.
...as assumed in paragraph~\ref{para:key}, the planner...
```

The second paragraph above is labelled `para:bind`; referring to it here resolves to paragraph 2. A fourth paragraph can then build on it explicitly:

4. Building on the assumption introduced in paragraph 2, the planner's tax τ^* restores constrained efficiency.

> Cross-references resolve inside environments too: this `notebox` cites paragraph 2 directly, confirming that `\para` labels behave like any other counter and that `\para` coexists with `notebox` without disturbing the count.

VIII Pencil Arrows

VIII.A Inline Directional Arrows

The `\jarrow*` commands produce small baseline-aligned pencil arrows. Follow them with `\ ` (backslash-space) to keep the following space, since the command gobbles it:

```
GDP growth \jarrowup\ accelerated in Q3.
\jarrowup[color=tomato]   % options via key=value
\jarrowr[len=0.6cm,lw=0.1cm]   % \jarrowr = \jarrowright
```

renders as (all four directions are available):

- Up: GDP growth ↑ accelerated in Q3.
- Down: Inflation ↓ fell below target.
- Right: The effect propagates → through the banking sector.
- Left: Demand shifted ← after the shock.

Custom options via key=value: ↑ (tomato), → (long, thick), ↓ (tight blue).

VIII.B Overlay Arrows with Marker Nodes

`\marker{name}{text}` wraps visible text in a named TikZ node. A subsequent overlay `tikzpicture` draws an `arrow`-style line to it. Compile **twice** for cross-picture references to resolve:

```
Consider the \marker{fisherA}{Fisher equation}:
\begin{tikzpicture}[tpstyle]
    \draw[arrow] ([yshift=-3pt]fisherA.south) to[bend right=20]
        +(0.2,-0.6) node[anchor=north west] {\hand links $r$ and $\pi$};
\end{tikzpicture}
```

renders as:

Consider the Fisher equation and the quantity theory:

links r and π links M and P

IX Package Options

This section documents the two options the package accepts at load time, passed in the optional argument of `\usepackage`: `wiggle=<n>` sets the amplitude of the pencil decoration, and `hidememo` suppresses all `\memo` output for a clean final build. Each is demonstrated below.

IX.A Wiggle

The pencil amplitude is controlled by the `wiggle=` package option:

- `\usepackage{jambronotes}` → default amplitude (0.4pt)
- `\usepackage[wiggle=0.3]{jambronotes}` → subtle
- `\usepackage[wiggle=1.5]{jambronotes}` → expressive

This document uses the default. Recompile with a different option to compare:

$\boxed{Y = \alpha + \beta X + \varepsilon}$ underline at default wiggle

IX.B Memo Suppression

Use `\usepackage[hidememo]{jambronotes}` when you want the same notes document without draft annotations. In that mode every `\memo` call expands to nothing, while the rest of the style remains unchanged.

X Front and Back Matter

X.A TOC Headings and the List of Boxes

The List of Boxes after the table of contents is produced by `\listofboxes`; every numbered box (created with `\boxentry`, Section III) adds an entry. `\tocheading` typesets an unnumbered heading in the TOC font, useful for labelling front-matter sections:

```
\tableofcontents
\listofboxes              % auto-titled "List of Boxes"
\tocheading{Supplementary Material}   % unnumbered heading
```

`\tocheading{Example Heading}` renders as:

Example Heading

X.B Appendix Numbering

`\backmatter` switches `\section` numbering to letters and restarts it at A (and re-prefixes hyperlink anchors so they stay unique), matching the appendix convention of the VAR Toolbox Handbook. Call it before an appendix; the section below uses it, so it is lettered A even though it follows Section X.

```
\backmatter
\section{Appendix: Quick Reference}
```

A Appendix: Quick Reference

This section follows Section X in the source but is lettered A, because `\backmatter` switched section numbering to letters and reset the counter.